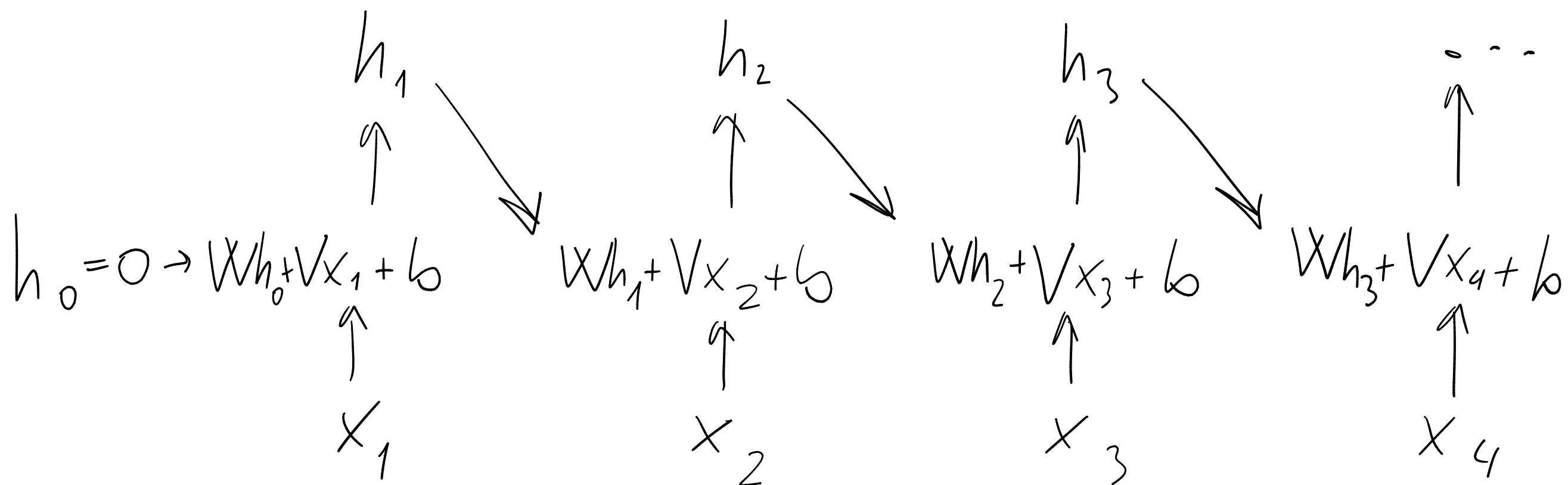


Recurrent Neural Networks

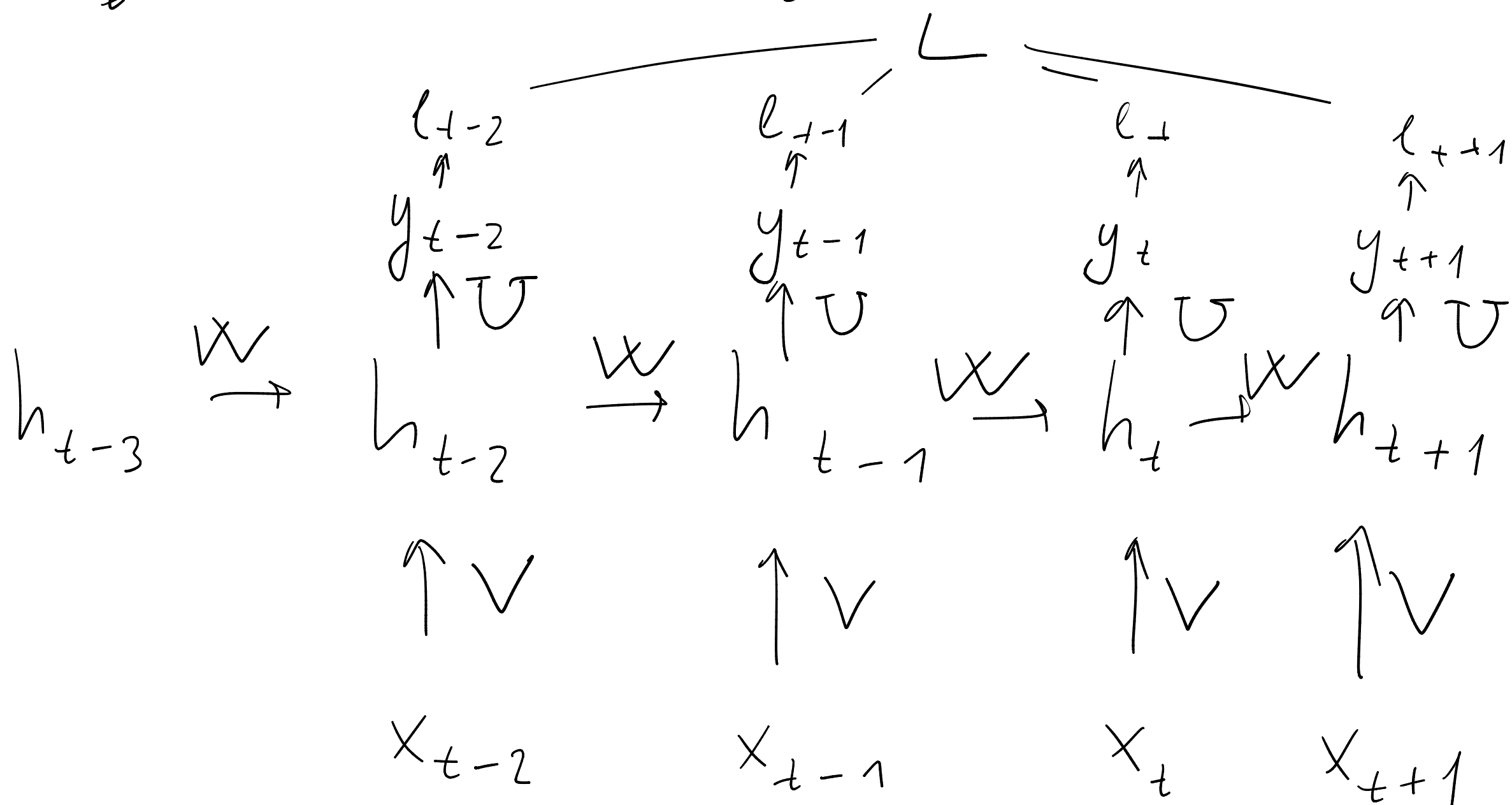


$$x_i \in \mathbb{R}^d$$

$$\begin{pmatrix} V & | & W \end{pmatrix} \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

h - hidden state

$$h_t = f(W h_{t-1} + V x_t + b)$$



$$L = \frac{1}{T} \sum_{t=1}^T \ell_t$$

$$\frac{dL}{dU} = \frac{1}{T} \sum_{t=1}^T \frac{d\ell_t}{dU} = \frac{1}{T} \sum_{t=1}^T \frac{d\ell_t}{dy_t} \frac{dy_t}{dU}$$

$$\frac{dL}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dy_t} \cdot \frac{dy_t}{dh_t} \cdot \frac{dh_t}{dW}$$

$$\frac{dh_t}{dW} = \frac{\partial h_t}{\partial W} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{dh_{t-1}}{dW} = \frac{\partial h_t}{\partial W} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial W} +$$

$$+ \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}} \cdot \frac{dh_{t-2}}{dW} = \sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \frac{\partial h_{t-j}}{\partial W}$$

$$\frac{dL}{dW} = \frac{1}{T} \sum_{t=1}^T \frac{dL_t}{dy_t} \frac{dy_t}{dh_t} \sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \frac{\partial h_{t-j}}{\partial W}$$

BPTT (Backpropagation through time)

$$\prod \frac{\partial h_{k+1}}{\partial h_k}$$

• $\left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| > 1$ — exploding gradients, divergence, NaNs

Gradient clipping $g \leftarrow \min \left(1, \frac{\text{thr}}{\|grad\|} \right) \cdot grad$

$g \leftarrow \text{clip}(grad, c, -c)$

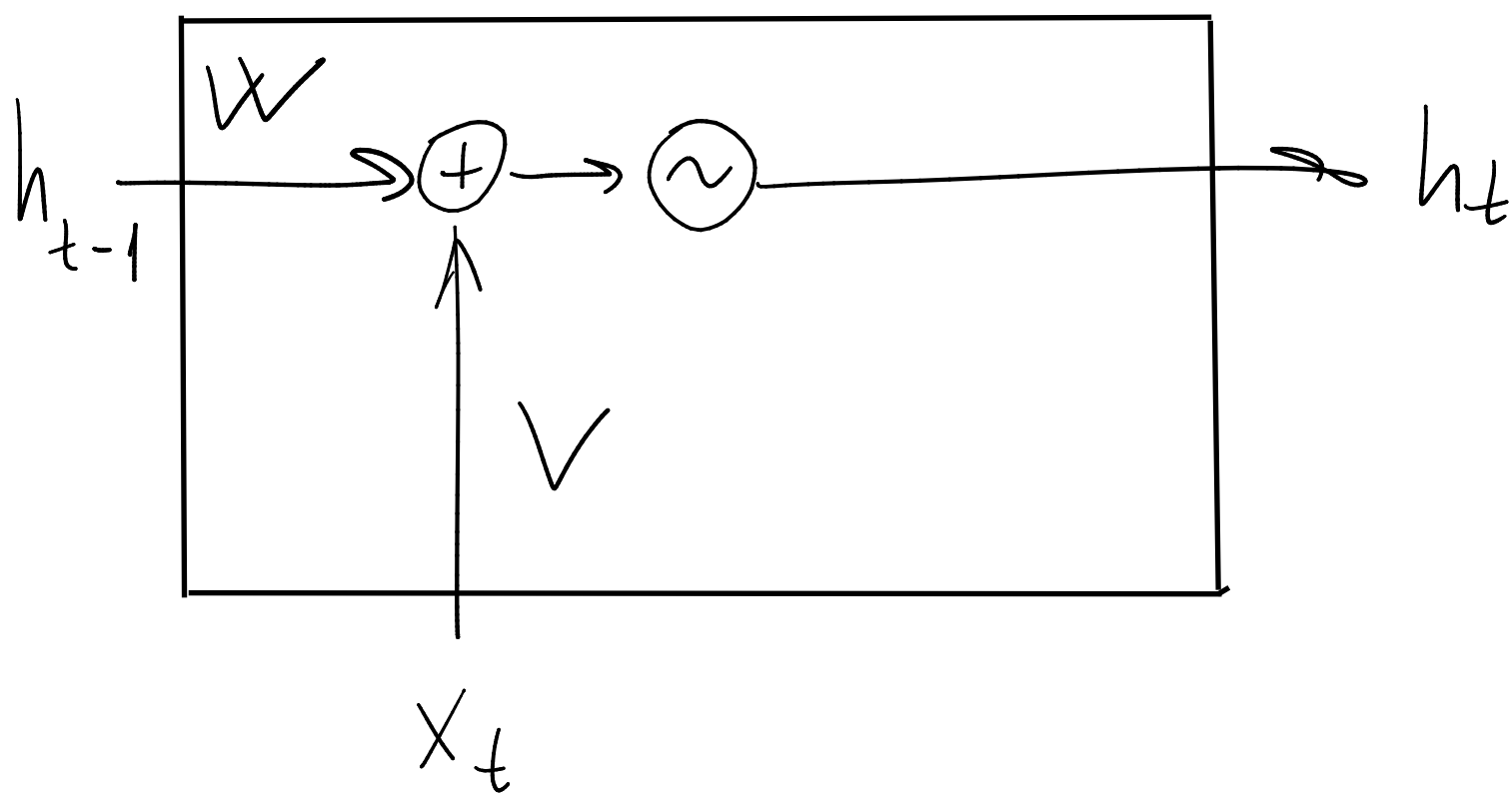
• $\left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| < 1$ — vanishing gradients

$$h_{k+1} = f(\underbrace{W h_k + V x_{k+1} + b}_{z_{k+1}})$$

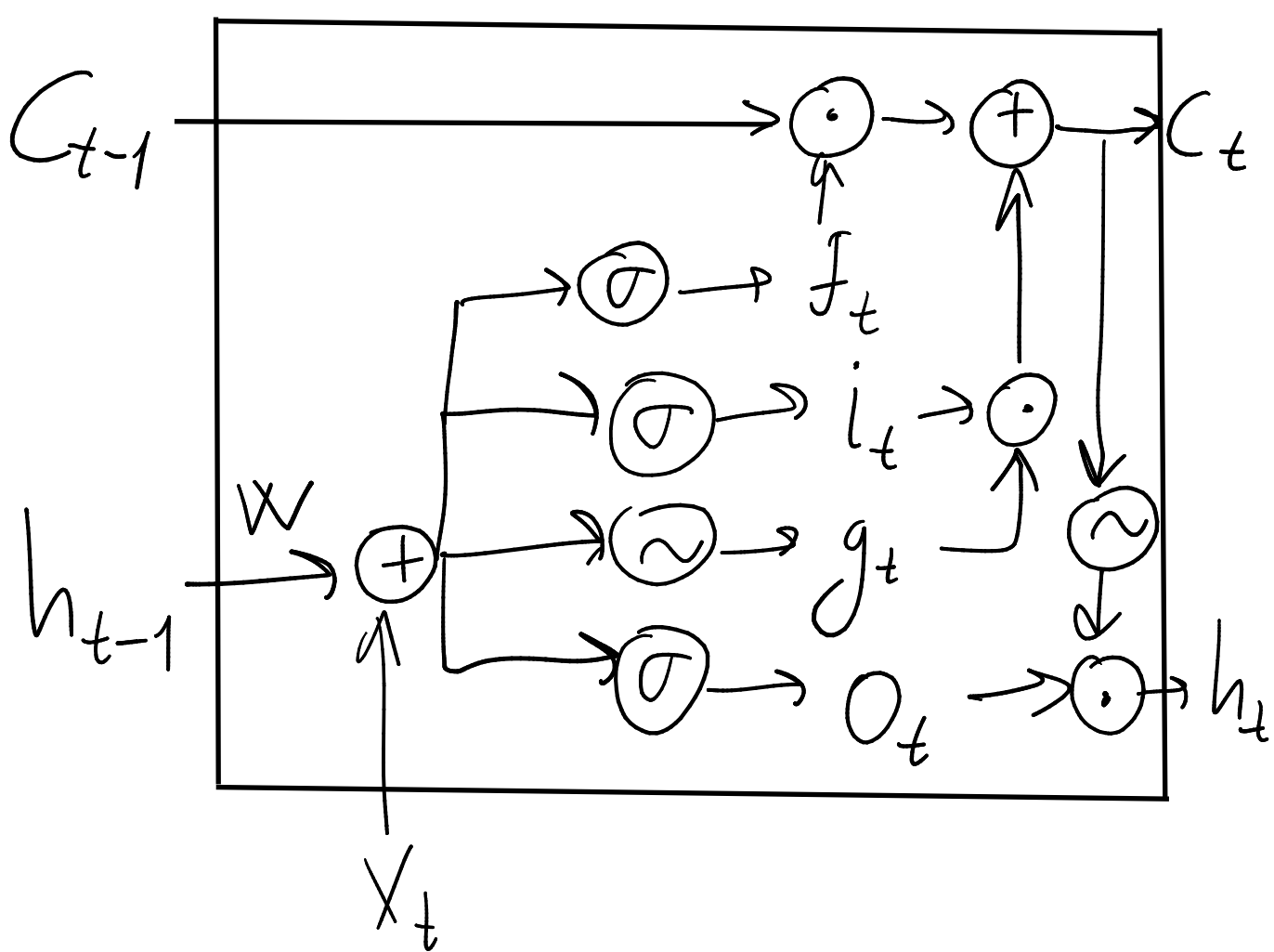
Init W with orthogonal matrix

$$\left\| \frac{\partial h_{k+1}}{\partial h_k} \right\| = \left\| \text{diag} \left(f'_{h_k}(z_{k+1}) \right) \right\| \cdot \|W\|$$

RNN



LSTM (long short term memory)



$$g_t = f(W_g h_{t-1} + V_g x_t + b_g)$$

$$i_t = \sigma(W_i h_{t-1} + V_i x_t + b_i)$$

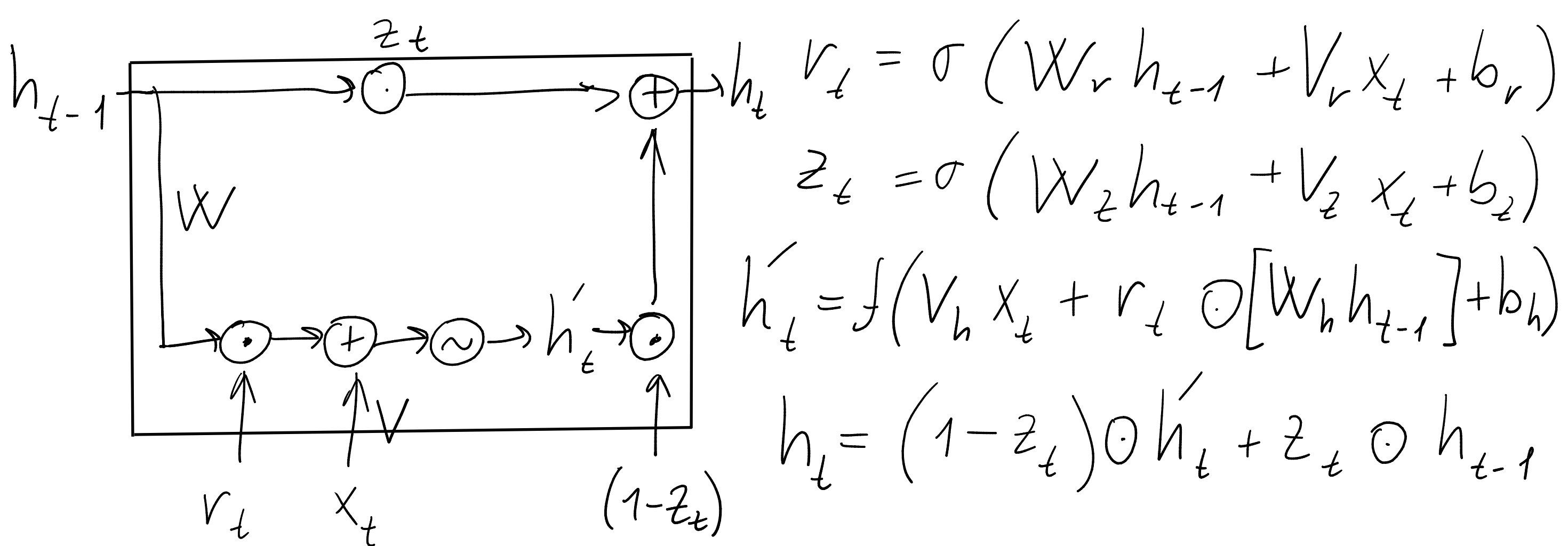
$$o_t = \sigma(W_o h_{t-1} + V_o x_t + b_o)$$

$$f_t = \sigma(W_f h_{t-1} + V_f x_t + b_f)$$

$$C_t = C_{t-1} \odot f_t + i_t \odot g_t$$

$$h_t = f(C_t) \odot o_t$$

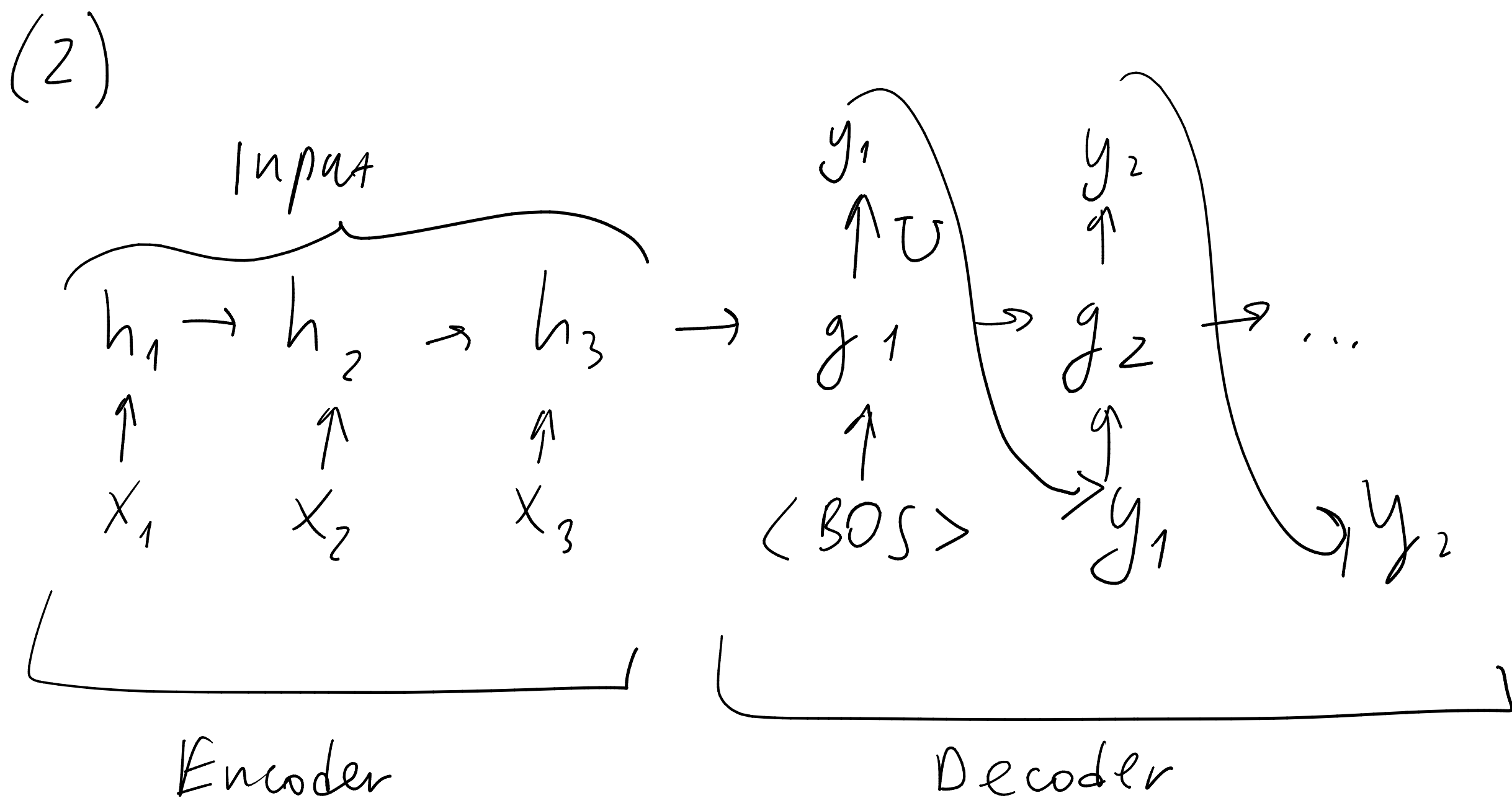
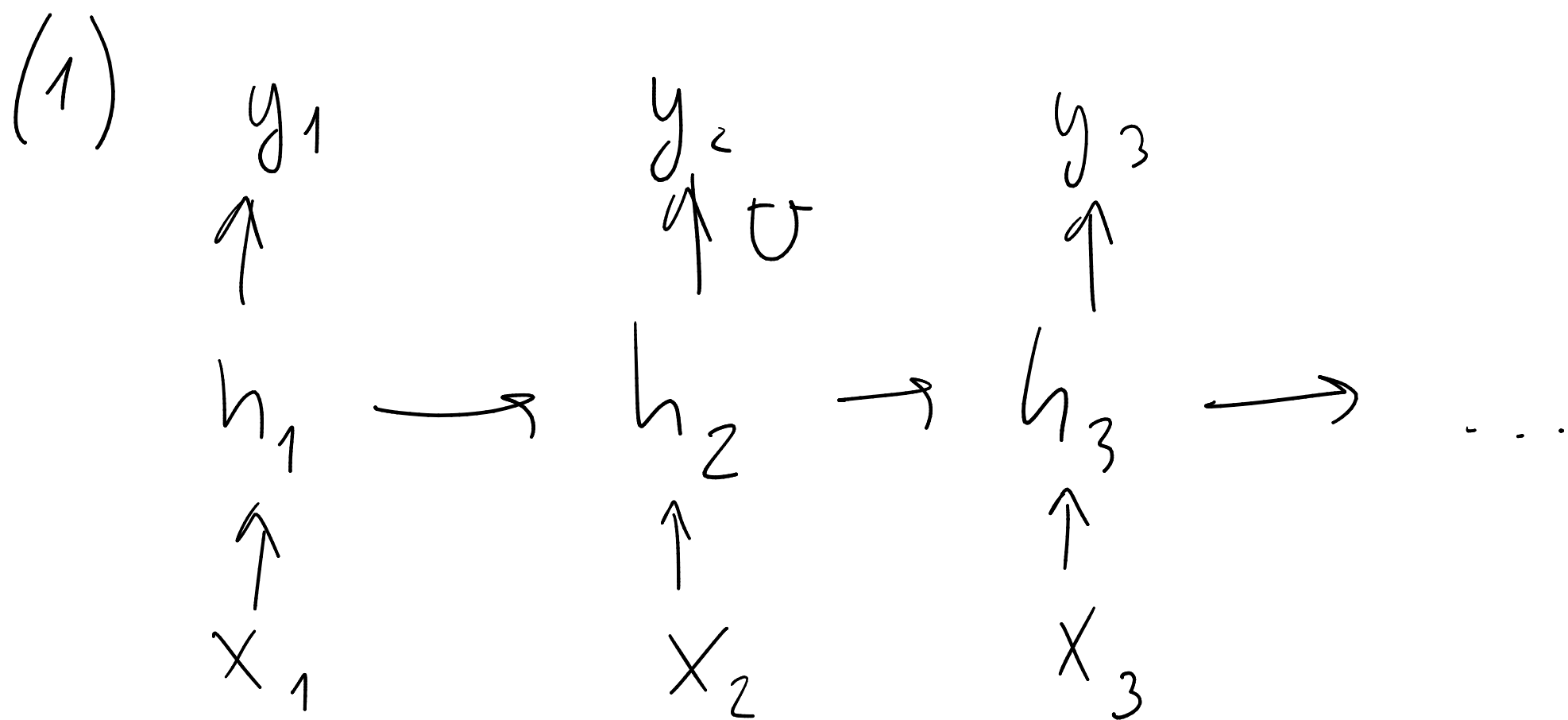
GRU (gated recurrent unit)



• Many-to-many (Seq-to-seq)

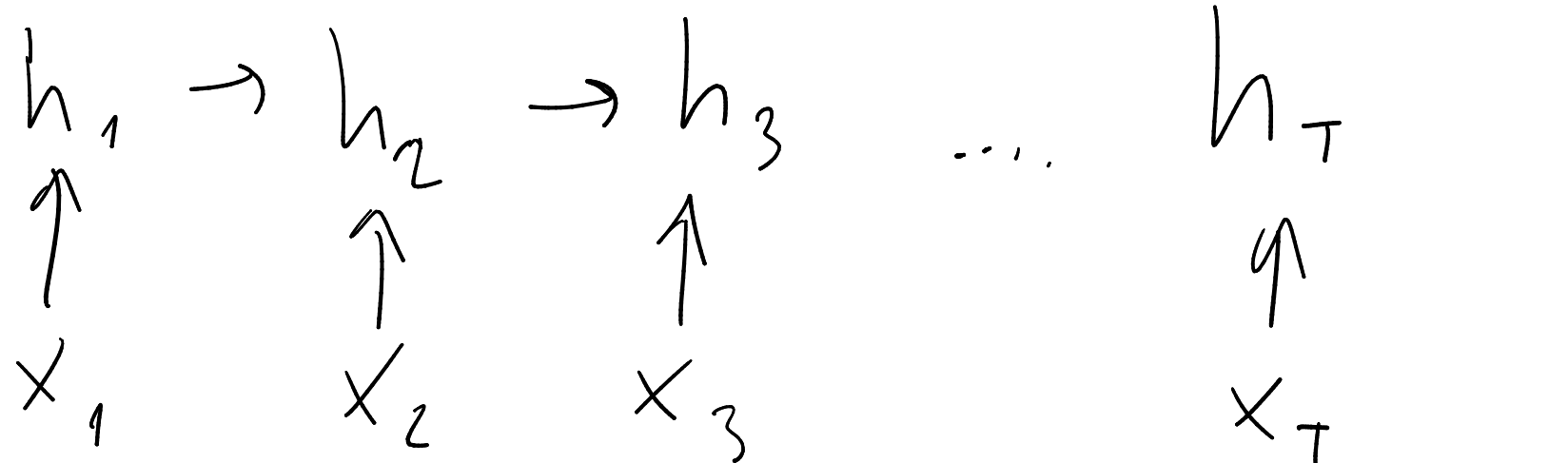
- Part of speech tagging (1)

- Machine translation (2)



- Many-to-one

- Text classification

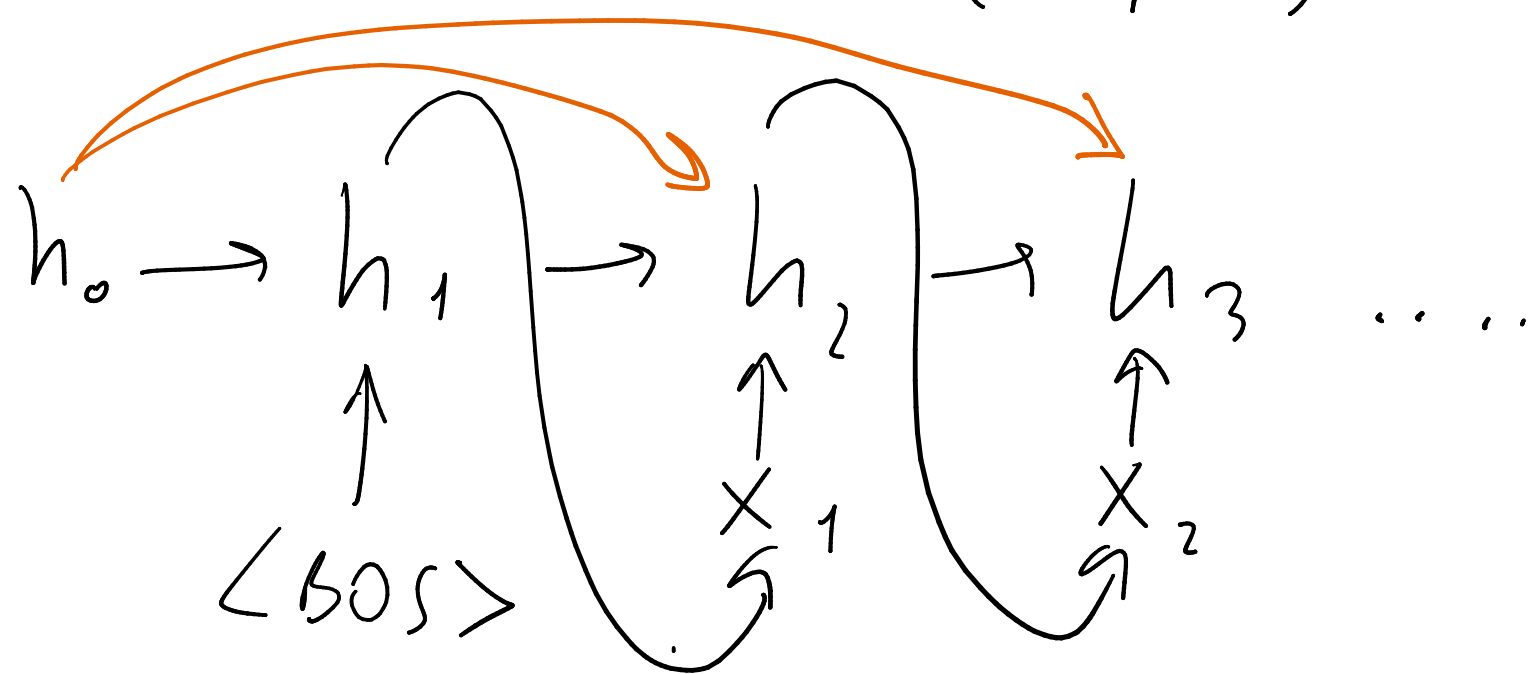


- One-to-many

- Conditional generation (Image captioning)

genre $\rightarrow$ music generation

$h_0 = \text{Encoder}(\text{Input})$ or $\text{Linear}(\text{Encoder}(\text{Input}))$

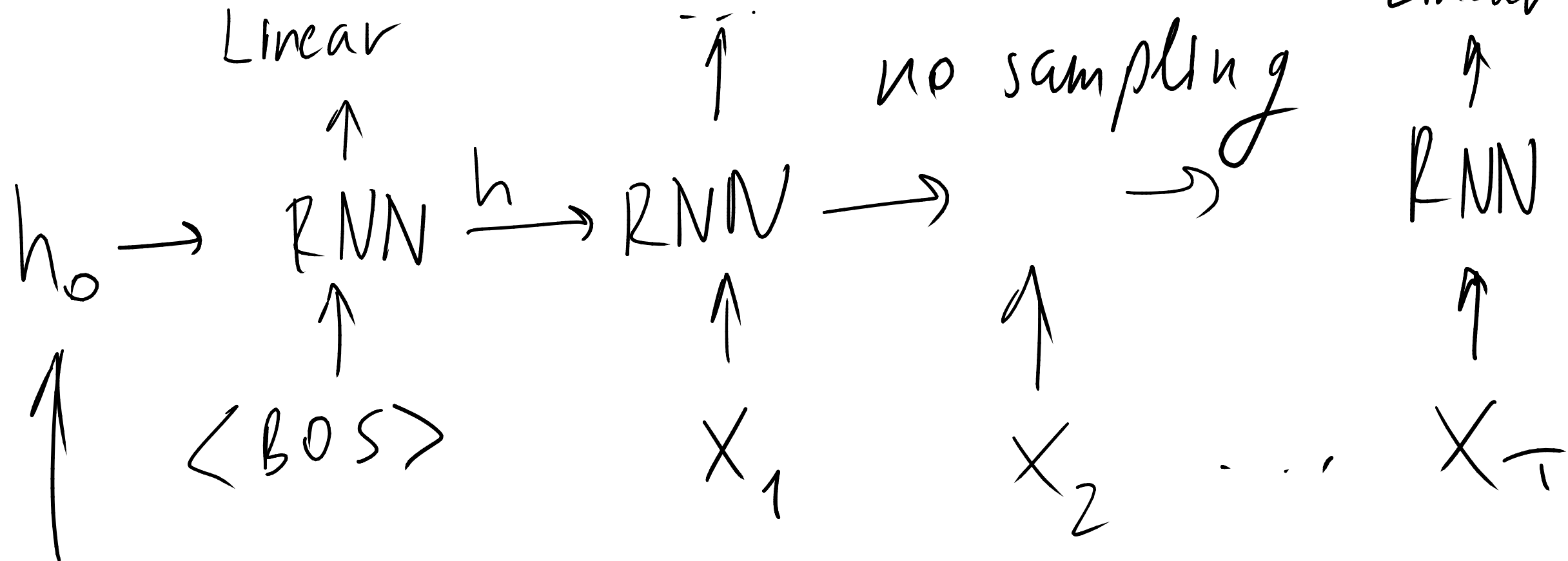
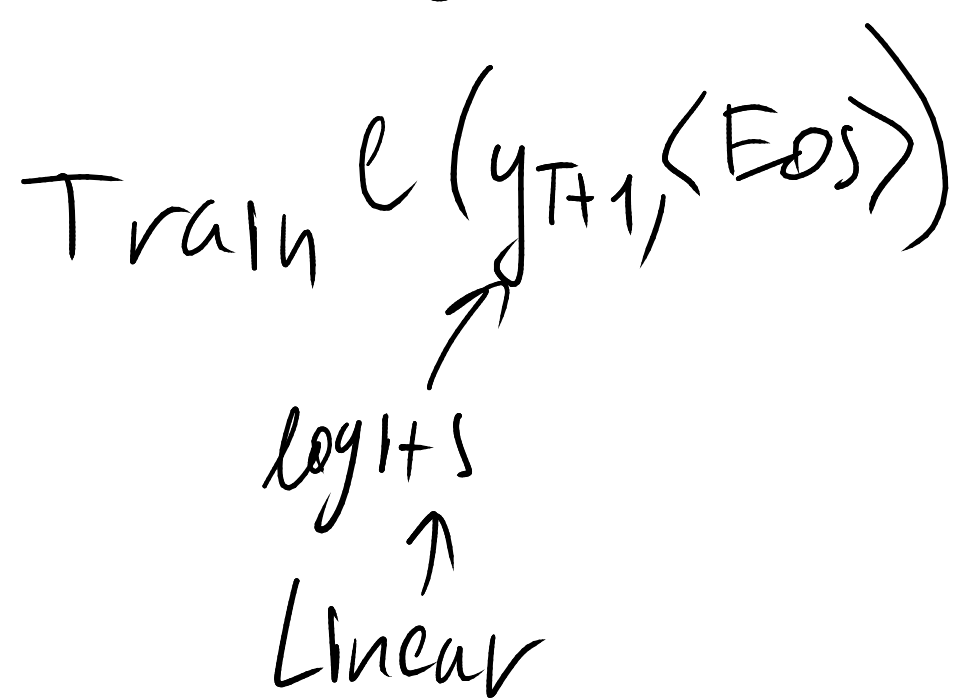
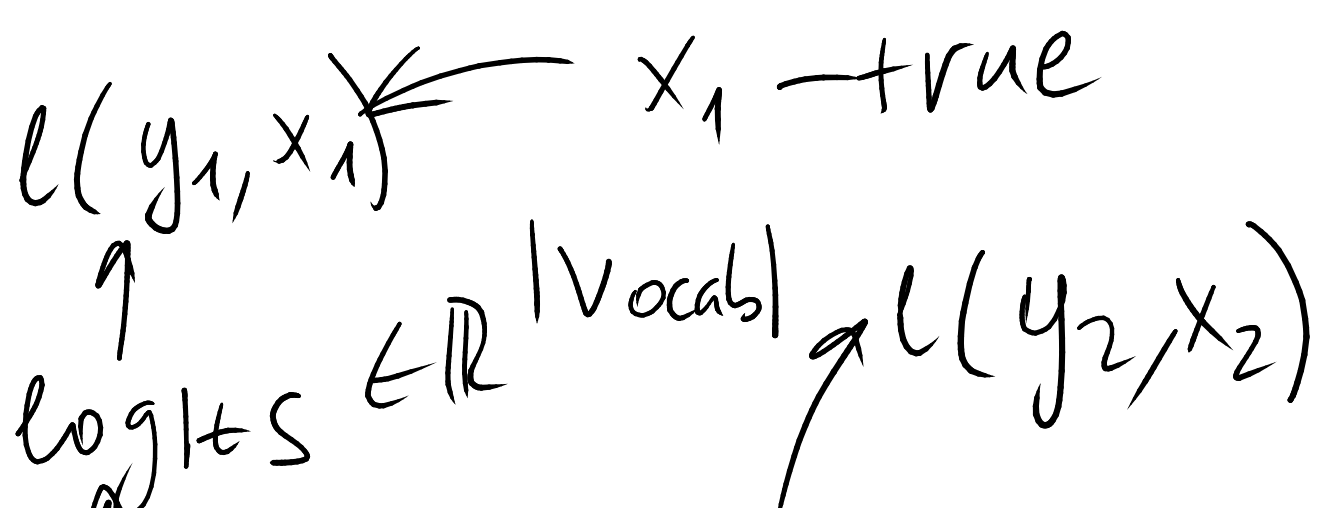
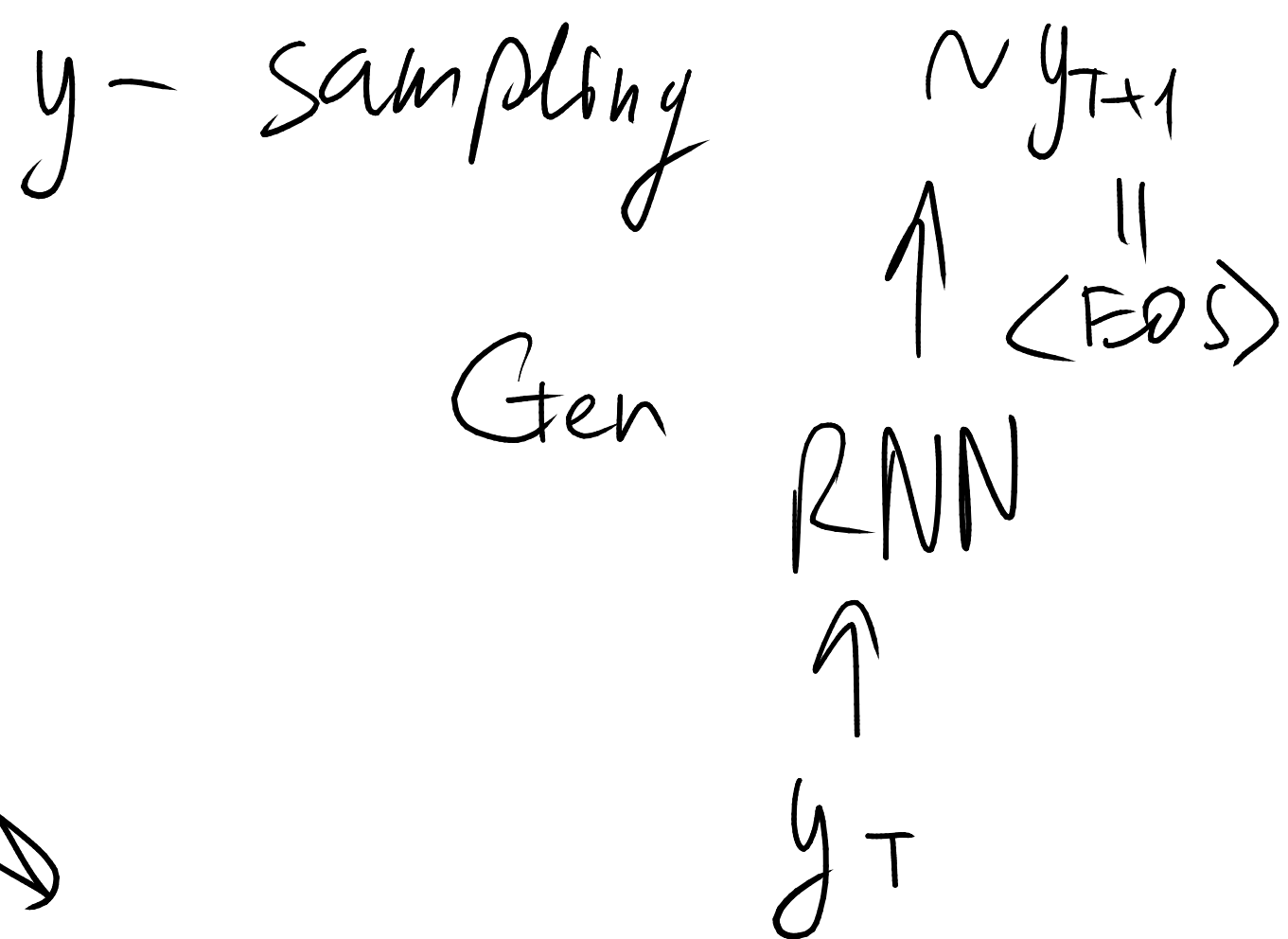
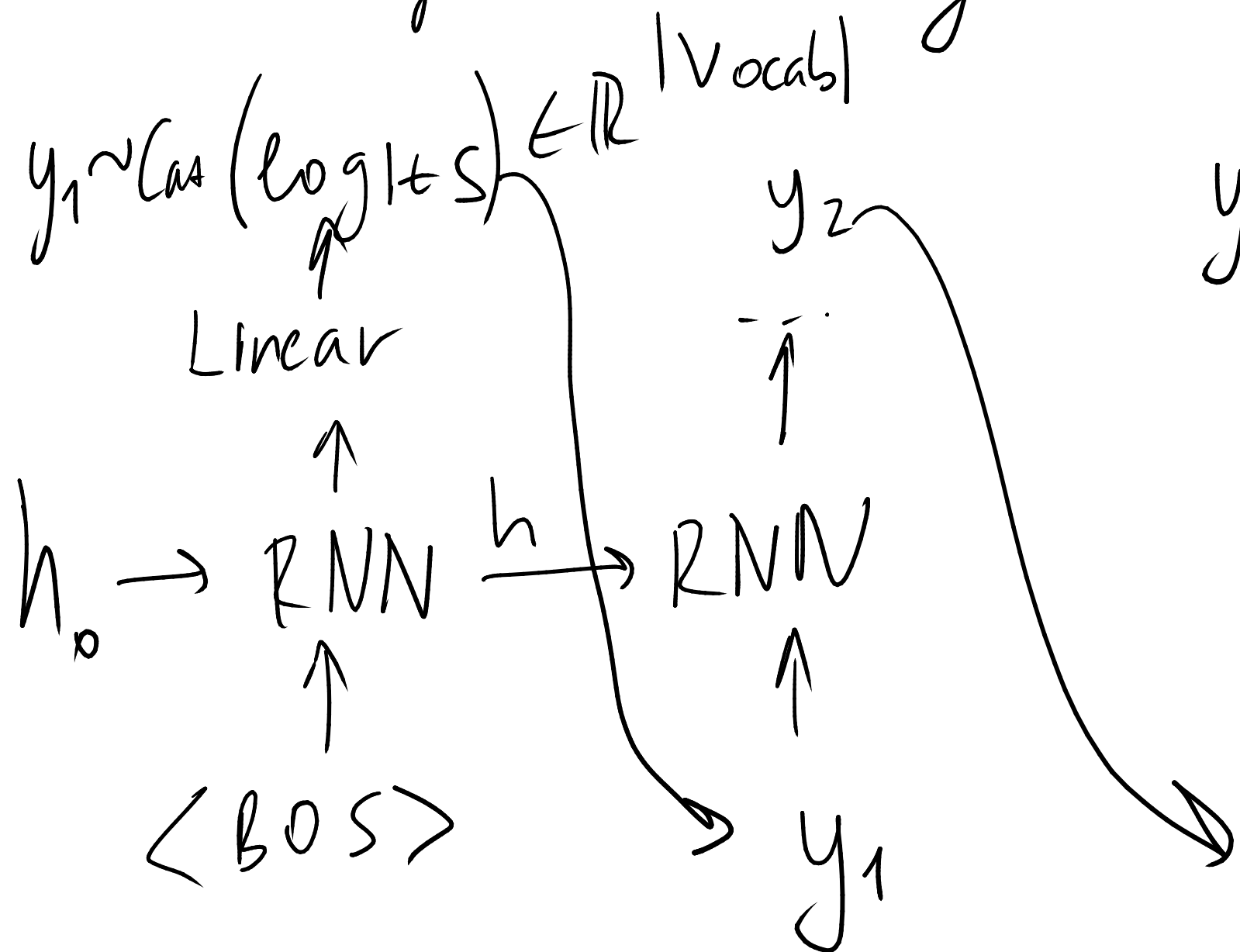


Bidirectional LSTM

Language Model

$$p(x) = p(x_1) p(x_2 | x_1) p(x_3 | x_1, x_2) \dots p(x_T | x_1, \dots, x_{T-1})$$

Autoregressive Modeling



May be from Encoder

θ - model params

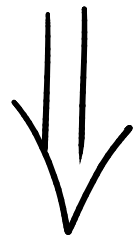
$$P_{\theta}(x_T | x_1, \dots, x_{T-1})$$

$$X = (x_0, \dots, x_L)$$

$$X_i = (x_{i1}, \dots, x_{iT})$$

$$p_{\theta}(X_i) = p(x_{i_0}) p_{\theta}(x_{i_1} | x_{i_0}) \dots p(x_{i_T} | x_{i_0}, \dots, x_{i_{T-1}})$$

$$x_{i_0} = \langle \text{bos} \rangle \quad \begin{matrix} \parallel \\ 1 \end{matrix}$$



$$-\sum_{t=1}^T \log p_{\theta}(x_{i_t} | x_{i_0} \dots x_{i_{T-1}}) \rightarrow \min$$

+ conditional